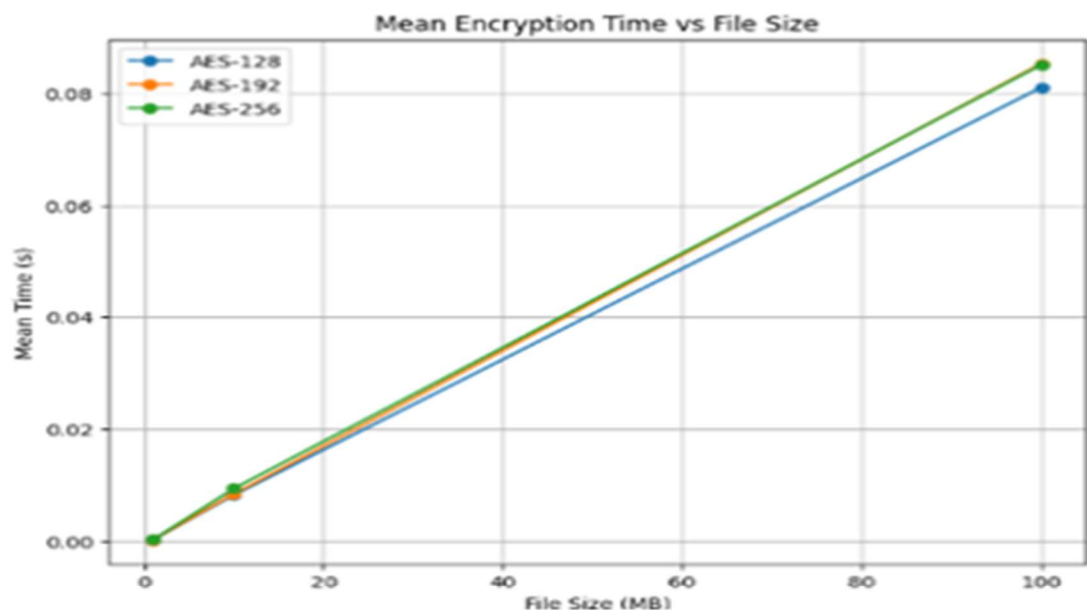
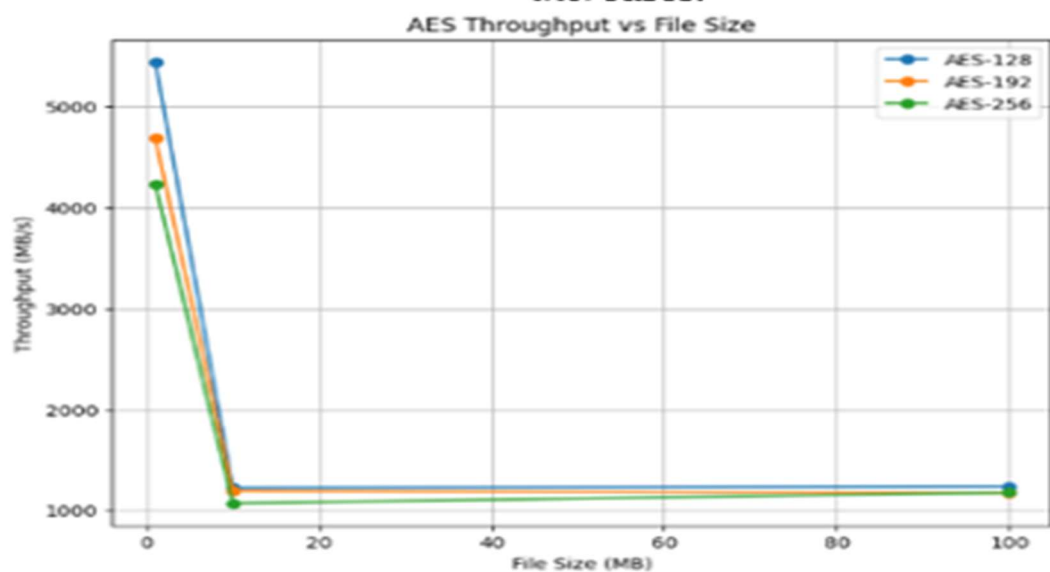




*Figure V.4: Mean AES encryption time (seconds) for AES-128, AES-192, and AES-256 at 1 KB, 10 MB, and 100 MB. The performance gap between key lengths narrows as file size increases.*



*Figure V.5: AES throughput (MB/s) by key length and file size. All three variants converge to approximately 1,170–1,234 MB/s at 100 MB, indicating the AES round-count penalty is negligible at scale.*